

ASEN 6044 Advanced State Estimation

Spring 2026

Assignment 3

Out: Tues 03/03/2026 (posted on Canvas + Gradescope)

Due: Thurs 03/26/2026, 11:59 pm (via Gradescope)

*For theory questions and derivations: you may consult reference materials (cite appropriately), but do not rely on symbolic math programs to find derivatives or integrals (e.g. use tables instead to do these analytically). For coding questions: use whatever programming environment you wish, i.e. Matlab, Python, R, Julia, etc. – unless otherwise indicated, you may use toolboxes or built-in functions in place of writing certain common routines for yourself (e.g. sampling random numbers, numerical optimization). What you turn in should be neatly prepared, and concisely address all parts of the questions provided below in a clearly organized manner. Be sure to precisely explain the logic of your solution (including key steps in any derivations), and appropriately comment your code (code should be in an appendix). **You may discuss your work with other students/groups, but submitted work must be your own.***

1. Consider a simultaneous state and parameter estimation problem, where the scalar random static state $x \sim p(x) = \mathcal{U}[1, 11]$ is measured with i.i.d. observations $y \sim p(y|x, z) = \mathcal{U}[x, (x + z)^2]$. Here, the measurement bias z is a Gamma distributed random variable independent of x such that

$$p(z) = \mathcal{G}(a, b) = \frac{1}{b^a \Gamma(a)} z^{a-1} e^{-\frac{z}{b}}, \quad (1)$$

where the shape parameter $a = 2$ and scale parameter $b = 2$. Suppose the following three measurements are taken in sequence,

$$y_1 = 91.56, \quad y_2 = 70.43, \quad y_3 = 108.67. \quad (2)$$

- (a) Use a grid-based approximation to plot the sequence of 2D joint posterior pdf surfaces $p(x, z|y_1)$, $p(x, z|y_1, y_2)$ and $p(x, z|y_1, y_2, y_3)$, for grid points on $x \in [1, 14]$ and $z \in (0, 30]$ of size $\Delta x = 0.05$ and $\Delta z = 0.05$. Comment on the results.
- (b) Use the grid approximation results from (a) to approximate the MAP and MMSE estimators for $x_0 = [x, z]^T$ given: y_1 only; y_1 and y_2 only; and all 3 observations. What are the values of the marginal data likelihoods $p(y_1)$, $p(y_1, y_2)$ and $p(y_1, y_2, y_3)$?
- (c) Approximate the posterior pdf $p(x, z|y_1, y_2, y_3)$ using Monte Carlo IS with 500 samples, using the prior as the proposal distribution $q(x, z; y_1, y_2, y_3)$. Use the resulting weighted sample set to compute the joint MMSE estimate for $x_0 = [x, z]^T$, and generate a 3D scatter plot to visualize the weighted particles (with vertical value and color of markers scaled by IS weights for different sampled x and z values, e.g. using Matlab's `scatter3` function, or something similar). How does the MMSE estimate from IS compare to the grid-approximated MMSE estimate?

2. Hidden Markov models (HMMs) are useful for devising approximations to the general Bayes filter when a stochastic dynamical system's states can be restricted to a finite domain. Consider the problem of localizing an object that moves and is observed stochastically in a bounded domain, e.g. an old jalopy moving on a long, windy, and unpaved road that is monitored by noisy unattended ground sensors. Let the jalopy's state be given by its scalar position, $x(k) \in [0, L]$, where L is the length of the road. Suppose the road is discretized into N_x grid cells with spacing Δx , and that the jalopy moves with some random constant velocity over time interval Δt between each time step $k = 0, 1, 2, \dots$. In this case, the random state variable $x(k)$ can be replaced by $\zeta(k) \in \{1, \dots, N_x\}$, which indexes a grid cell for $x(k)$ at time k and whose dynamics are well-approximated by an HMM¹.

For grid cells $i, j \in \{1, \dots, N_x\}$, let the HMM conditional state index transition probabilities $P(\zeta_{k+1} = i | \zeta_k = j)$ (which approximately encode the spatially and temporally discretized system dynamics) be defined by the following time-invariant parameters,

$$P(\zeta_{k+1} = i | \zeta_k = j) = \begin{cases} 0.3, & \text{if } i = j \text{ and } i \in \{2, \dots, N_x - 1\} \\ 0.5, & \text{if } i = j + 1 \text{ and } i \in \{3, \dots, N_x\} \text{ and } j \geq 2 \\ 0.2, & \text{if } i = j - 1 \text{ and } i \in \{1, \dots, N_x - 2\} \text{ and } j \leq N_x - 1 \\ 0.1, & \text{if } i = j = 1 \\ 0.1, & \text{if } i = j = N_x \\ 0.9, & \text{if } i = 2 \text{ and } j = 1 \\ 0.9, & \text{if } i = N_x - 1 \text{ and } j = N_x \\ 0, & \text{otherwise.} \end{cases}$$

The Chapman-Kolmogorov equation for recursive probabilistic state prediction can thus be approximated in terms of the marginal probability mass distribution over ζ_{k+1} ,

$$P(\zeta_{k+1} = i) = \sum_{j=1}^{N_x} P(\zeta_{k+1} = i | \zeta_k = j) P(\zeta_k = j).$$

Finally, suppose that an unattended ground sensor with resolution $2r$ provides noisy grid cell index observations $z_k = i \in \{1, \dots, N_x\}$ for the jalopy conditioned on $\zeta_k = j$, as given by the probabilistic observation likelihood

$$P(z_k = i | \zeta_k = j) = \mathcal{U}_{i|j}[a(r), b(r)], \\ a(r) = \max(1, j - r), \quad b(r) = \min(N_x, j + r),$$

For this problem, assume $r = 25$, $L = 1000$ m, $\Delta x = 10$ m, $\Delta T = 5$ s, and $N_x = 100$.

a. Suppose the vector $\rho_{k+1}^- \equiv [P(\zeta_{k+1} = 1), \dots, P(\zeta_{k+1} = N_x)]^T$. Find the square matrix T such that $\rho_{k+1}^- = T\rho_k^-$, and determine whether a *stationary distribution* ρ_∞^- exists for the system, i.e. such that $\rho_\infty^- = T\rho_\infty^-$. (**Hint:** If ρ_∞^- exists, what would be special about the relationship between ρ_∞^- and T ?)

¹e.g., see: N. Ahmed, D. Casbeer, Y. Cao, and D. Kingston, "Multi-target Localization on Road Networks with Hidden Markov Rao-Blackwellized Particle Filters," AIAA Journal of Aerospace Information Systems, Vol. 14, No. 11 (2017), pp. 573-596.

b. Suppose ρ_0^- is such that $P(\zeta_0 = 3) = P(\zeta_0 = 19) = P(\zeta_0 = 35) = \frac{1}{3}$. Derive an expression in terms of ρ_0^- which gives the predicted grid index distribution ρ_k^- at some future time step $k > 0$, and plot ρ_k^- vs. ζ for $k = 10, 50, 180, 300$ and 600 time steps (e.g. using the ‘barplot’ function in Matlab). If ρ_∞^- exists, approximately how long does it take for ρ_k^- to get ‘close to’ ρ_∞^- ?

c. Use the prior in part (b) and the measurement data log ‘zMeasHist’ posted in the file ‘asen6044hw3_problem2.mat’ to implement the exact Bayes filter, i.e. compute a recursive posterior distribution $\rho_k^+ \equiv [P(\zeta_k = 1|z_{1:k}), \dots, P(\zeta_k = N_x|z_{1:k})]^T$ to solve the filtering problem at each time step in the data log (the first value of ‘zMeasHist’ corresponds to $z(0)$, which is deliberately set to NaN). Turn in plots for ρ_k^+ vs. $\zeta(k)$ at time steps $k = 7, 50, 180, 300$, and 350 ; report the corresponding MMSE and MAP estimates for $\hat{\zeta}_k^+$ at each of these specified time steps, and compare/discuss the results with respect to the ground truth ζ_k values provided in the array ‘zetaHist’ (where the initial value corresponds to the true $\zeta(0)$ initial condition).

d. Repeat part c, except this time use a bootstrap SIR particle filter (PF) approximation with $N_s = 500$ particles. Note that in this case particles will simulate a trajectory over the discrete HMM states $\zeta_{0:N_z}$ (where N_z is the number of measurements taken). For plots, approximate the posterior for ρ_k^+ vs. $\zeta(k)$ at the same specified time steps as in (c), using the weighted sum of delta functions corresponding to the PF’s representation of the posterior at time k . How do the posteriors and MMSE and MAP estimates derived from the SIR PF compare to those obtained using the exact Bayes filter from part (c) at the specified times?

Challenge Questions (For Fun) Attempt as many of these as you like, in any order you wish – **extra credit *may* be awarded for exceptionally good and complete answers (provided rest of assignment is completed in timely manner), but in any case these are optional and for your own edification.**

C1. Use importance sampling to find a MAP estimate of x_0 for challenge problem C1 of HW 2, using the same data log as provided in that problem. Explain the set up of your approach and give the relevant implementation details, along with relevant plots and analysis of the results. How does your answer compare to the answers obtained via the grid method and numerical optimization method?